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Project Proposal

Title: Development and Validation of a Multiplex Spatial PD-L1 Scoring System (CPS++) Using Model Context Protocol (MCP), AI-Agent, and CNN Image Classification for Triple-Negative Breast Cancer

- The project targets **Triple-Negative Breast Cancer (TNBC)**, which urgently needs reliable biomarkers for immunotherapy response.
- It introduces **CPS++**, an advanced version of the Combined Positive Score (CPS), to improve PD-L1 scoring accuracy.
- **Multiplex Spatial** emphasizes integrating PD-L1 staining with tumor–immune spatial context, beyond simple cell counts.
- The **Model Context Protocol (MCP)** is used to manage structured integration of multimodal data (image, clinical, molecular).
- **AI-Agent** acts as an orchestrator, automating tasks such as dataset selection, preprocessing, and feature fusion.
- **CNN-based image classification** provides robust tumor, immune, and stromal segmentation from histopathology and multiplex imaging.
- MCP ensures that contextual information (cell type, location, interaction) is preserved for downstream AI-Agent analysis.
- The **CPS++ algorithm** combines CNN-derived PD-L1 signals with spatial interaction indices into a unified biomarker score.
- This system enhances reproducibility and objectivity, overcoming variability in manual PD-L1 scoring by pathologists.
- Ultimately, the title conveys a **novel AI-Agent powered framework with MCP-driven data contextualization** to guide TNBC immunotherapy.

Background and Rationale

Triple-Negative Breast Cancer (TNBC) accounts for ~15–20% of breast cancers and is characterized by aggressive behavior and lack of targeted therapies. Immunotherapy using PD-1/PD-L1 inhibitors has shown promise, particularly in PD-L1 positive TNBC subsets. Current PD-L1 scoring methods (IC and CPS) suffer from inter-observer variability and fail to fully incorporate spatial tumor–immune interactions. We propose a Multiplex Spatial PD-L1 Scoring System (CPS++), integrating multiplex IHC/imaging, spatial analysis, and computational pathology through AI Agent orchestration, CNN-based image classification, and MCP protocol–driven multiplexing.

Objectives

1. Develop a CPS++ scoring algorithm combining PD-L1 tumor proportion, immune infiltration, and spatial distribution metrics. 2. Integrate an AI Agent for automated orchestration of image preprocessing, scoring, and validation. 3. Implement CNN-based classification for accurate tumor and immune cell segmentation. 4. Use MCP (Model Context Protocol) to handle multi-channel/multiplex imaging datasets. 5. Validate CPS++ using publicly available TNBC datasets (TCGA, TCIA, HTAN) and clinical trial cohorts (IMpassion130, KEYNOTE-355). 6. Benchmark CPS++ against existing PD-L1 scoring systems (IC and CPS) for predicting

progression-free survival (PFS) and overall survival (OS).

Methodology

Data Sources: - Discovery Cohort: TCGA-BRCA TNBC cases, TCIA whole-slide images, HTAN multiplex datasets. - Validation Cohorts: IMpassion130, KEYNOTE-355, real-world TNBC cohorts. CPS++ Scoring Pipeline with AI Agent and MCP Protocol:

1. Image Acquisition & Preprocessing: - H&E;, PD-L1 IHC, and multiplex IF data collected under MCP protocol. - AI Agent automates quality control, channel alignment, and metadata standardization.

2. Cell Segmentation & Classification: - CNNs trained on annotated TNBC images classify tumor, immune, and stromal compartments. - AI Agent supervises adaptive model retraining across cohorts.

3. Feature Extraction: - PD-L1 expression on tumor and immune cells. - Immune density and spatial interaction indices (nearest-neighbor, Ripley's K).

4. Algorithm Development: - CPS++ = f(Tumor PD-L1%, Immune PD-L1%, Immune Density, Spatial Index). - AI Agent integrates CNN outputs, multiplex readouts, and spatial features. 5. Statistical Analysis: - Kaplan–Meier survival, Cox regression, ROC/AUC, C-index benchmarking.

Layered Architecture

A. Data & Context Layer

- **Raw Data Repositories:** WSIs (H&E, PD-L1 IHC, mIF), clinical tables, assay metadata.
- **MCP Context Store:** Unified schema linking image tiles, cell masks, markers, patient/visit, therapy, outcomes, lab IDs, timestamps, and provenance.
- **Feature Store:** Precomputed tile/cell features (embeddings, PD-L1 intensities, densities, spatial metrics).
- **Model Registry:** Versioned CNNs, spatial models, CPS++ scoring configs, thresholds.

B. Processing & Analytics Layer

- **Preprocessing Service:** Tiling, color normalization, stain deconvolution, channel registration (for multiplex), QC.
- **Segmentation & Phenotyping (CNNs):** Tumor/immune/stroma segmentation; cell detection; PD-L1 quantification per compartment.
- **Spatial Analytics Engine:** Tumor–immune interface, proximity (kNN), Ripley's K/L, Moran's I, cluster density, boundary heatmaps.
- **CPS++ Scoring Engine:** Combines CPS inputs + spatial indices → composite CPS++; calibration and cutoffs per assay.

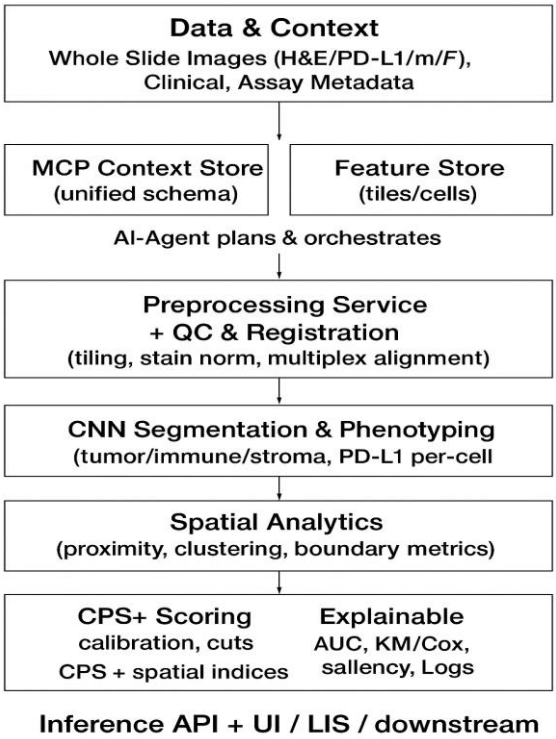
C. Orchestration & Intelligence Layer

- **AI-Agent Orchestrator:** Plans tasks, calls tools, monitors quality, adapts thresholds, schedules retraining; interacts via MCP to fetch/write context.
- **Validation & Stats:** ROC/AUC, C-index, KM/Cox models; drift detection; cross-site reproducibility checks.
- **Explainability & Reports:** Per-tile attributions, saliency for PD-L1 regions, spatial maps, audit logs.

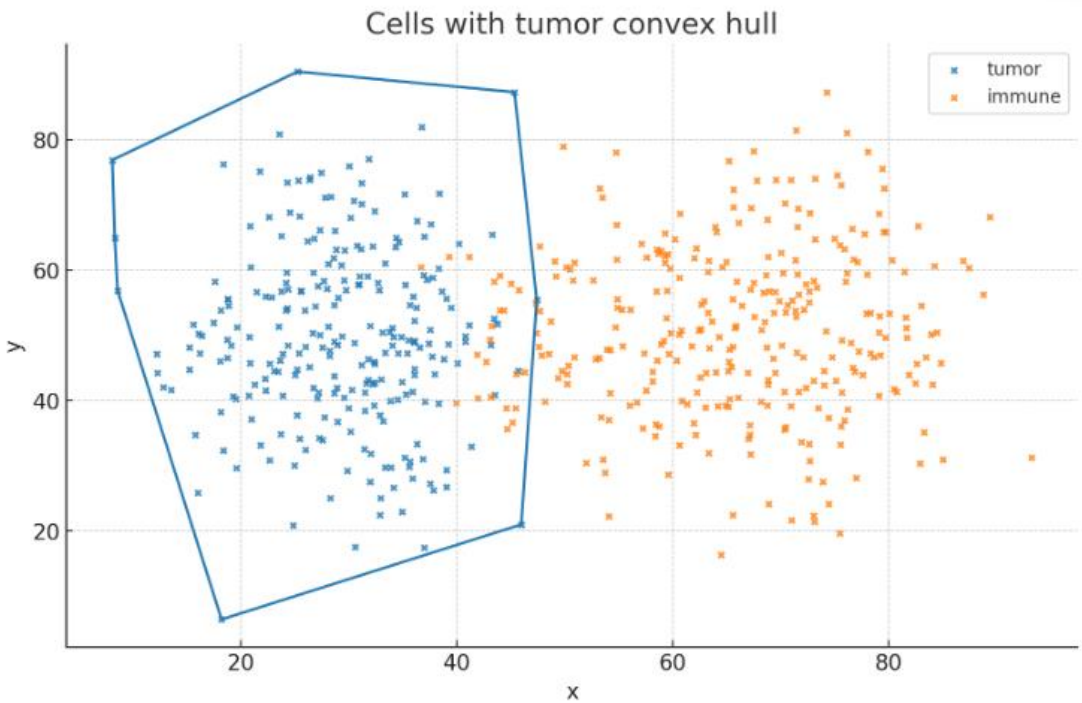
D. Serving & Access Layer

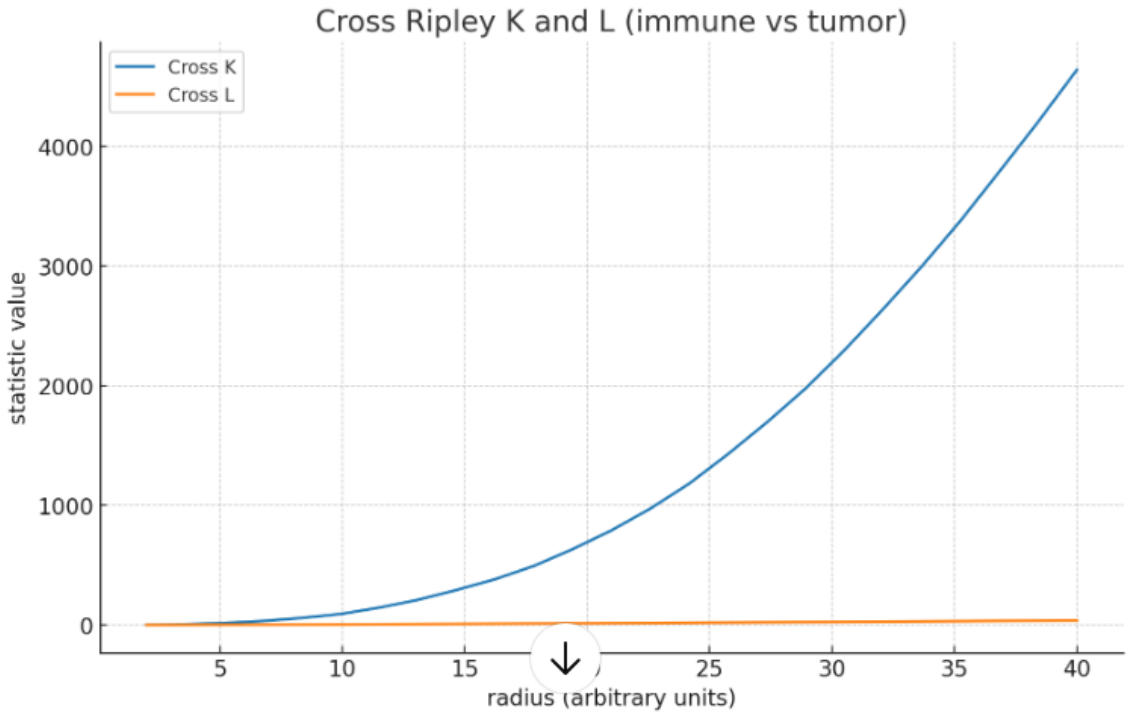
- **Inference API (REST/gRPC):** "Score case → return CPS++ + QC flags + explanations."
- **Reviewer UI (optional):** Slide viewer overlays (masks, hotspots, boundary bands), score breakdown, QC dashboard.
- **Security & Compliance:** PHI handling, access controls, audit trails, dataset use governance.

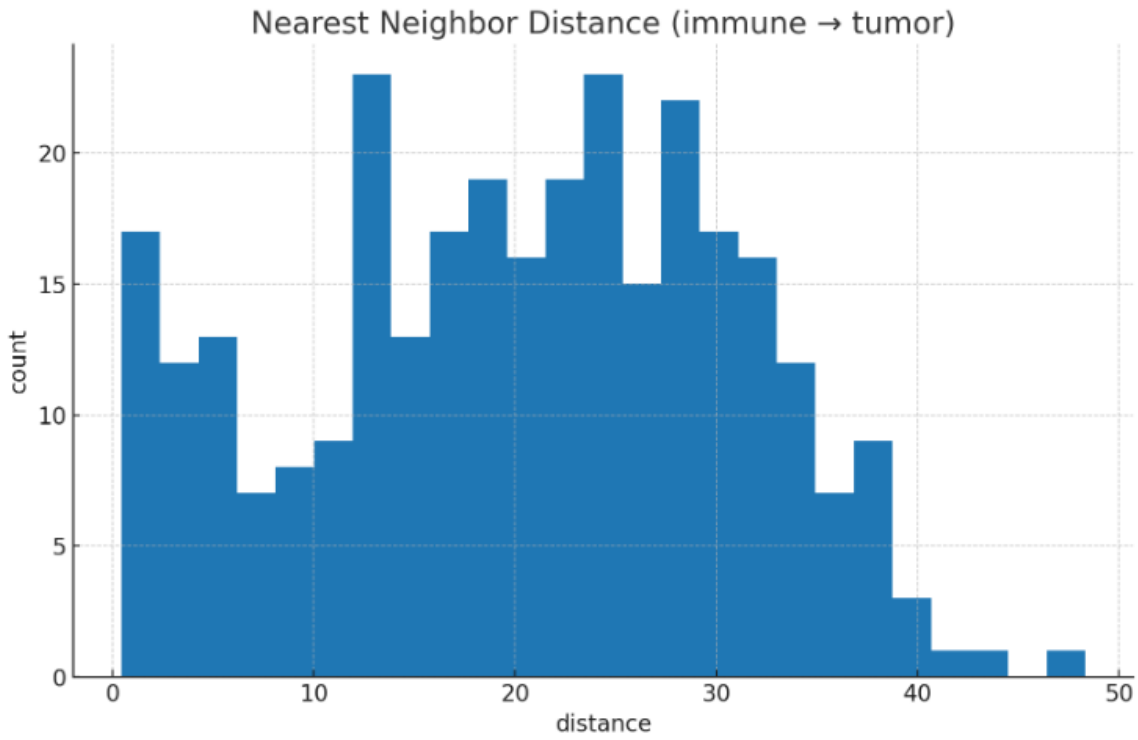
High-Level Dataflow Diagram



Tumor–immune interface, proximity (kNN), Ripley's K/L, Moran's I, cluster density, boundary heatmaps







Moran's I summary

1. Standard CPS (baseline)

The **FDA-approved CPS** for PD-L1 is:

$$CPS = \frac{\#(\text{PD-L1+ tumor cells} + \text{PD-L1+ lymphocytes} + \text{PD-L1+ macrophages})}{\#(\text{viable tumor cells})} \times 100$$

- Strength: Simple, widely adopted.
- Limitation: Ignores **spatial relationships** (immune infiltration pattern, tumor–immune interface, clustering).

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2. Proposed CPS++

Composite CPS++ = CPS × Spatial Interaction Modifier (SIM)

Where **SIM** is derived from normalized spatial indices:

- **Tumor–immune proximity (kNN)**
  - Closer immune–tumor distances increase SIM.
- **Interface density** (immune cells per μm of tumor boundary)
  - Higher boundary contact = stronger interaction.
- **Ripley's L (cross-type clustering)**
  - Captures tumor–immune co-localization beyond chance.
- **Moran's I (PD-L1 autocorrelation)**
  - High values → localized PD-L1 "hot spots."
- **Immune cluster density near boundary**
  - Weighted by PD-L1 status.

3. Normalization & Weighting

Each index s_j is normalized to 0–1 across the cohort:

$$\tilde{s}_j = \frac{s_j - \min(s_j)}{\max(s_j) - \min(s_j)}$$

Spatial Interaction Modifier (SIM):

$$SIM = \sum_{j=1}^{m} w_j \tilde{s}_j \quad \text{with} \quad \sum w_j = 1$$

Example weights (can be tuned by Cox/logistic regression):

- Proximity (kNN): 0.3
- Interface density: 0.3
- Ripley's L: 0.2
- Moran's I: 0.1
- Boundary cluster density: 0.1

4. Final Composite CPS++

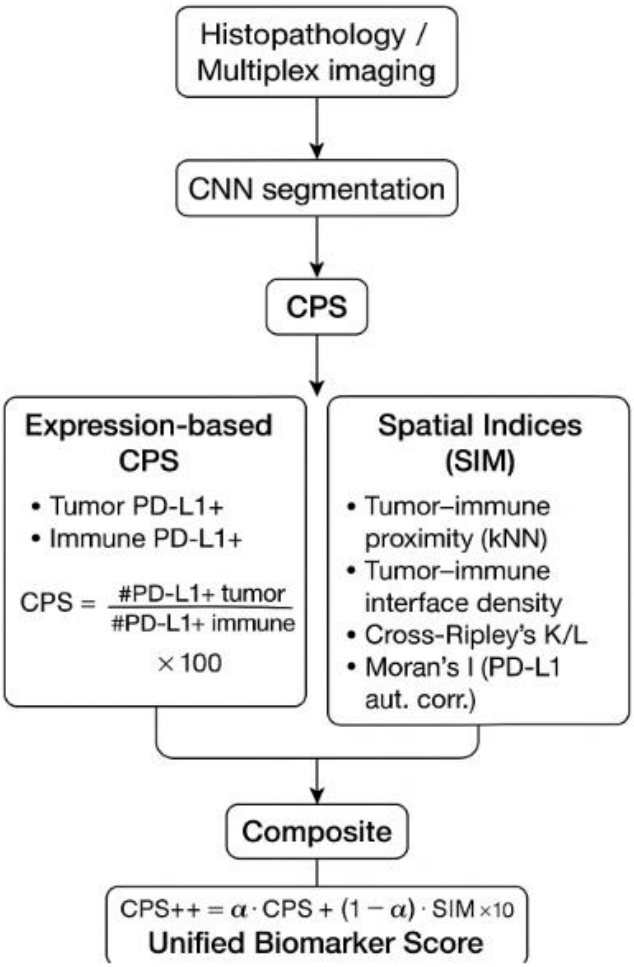
$$CPS^{++} = \alpha \cdot CPS + (1 - \alpha) \cdot SIM \times 100$$

- α balances expression (CPS) vs. spatial context.
- Example: $\alpha = 0.7$ (expression-dominant, but spatial adds nuance).

5. Interpretation

- **High CPS++** → strong PD-L1 expression **and** strong immune engagement at tumor boundary.
- **Low CPS but high SIM** → immune cells cluster near tumor, but PD-L1 expression is low → potential for alternative checkpoint pathways.
- **High CPS but low SIM** → PD-L1+ cells exist, but spatial exclusion of immune cells (immune desert).

Flow Diagram



Expected Outcomes

- A validated CPS++ biomarker integrating CNN-driven cell classification, multiplex MCP-based imaging, and AI Agent orchestration. - Superior predictive power over current PD-L1 scoring methods in TNBC immunotherapy. - Open-source AI Agent framework for reproducible TNBC biomarker research. - Foundation for a patentable diagnostic biomarker and computational system.

Timeline (5 months)

Phase	Duration	Activities
Phase 1: Data Collection	1 months	Acquire TCGA, TCIA, HTAN data; trial data requests
Phase 2: Algorithm Development	2 months	CNN training, AI Agent orchestration, CPS++ pipeline build
Phase 3: Discovery Validation	1 months	Validate on TCGA/HTAN datasets using MCP protocol
Phase 4: Clinical Validation	0.5 months	Apply CPS++ to IMpassion130, KEYNOTE-355 datasets
Phase 5: Reporting & IP Filing	0.5 months	Draft manuscript, file patent, prepare grant proposals

Deliverables

- CPS++ computational pipeline integrating AI Agent, CNN, and MCP protocol. - Benchmarking results vs IC and CPS scoring. - Manuscript for submission to high-impact oncology/AI journals. - Provisional patent application. - Conference abstracts (ASCO, SABCS, AACR).

Resources and Collaborations Needed

- Bioinformatics support (AI/ML developers, CNN experts, image analysts). - Pathology expertise for TNBC biopsies and PD-L1 IHC scoring. - Data access collaborations (Roche, Merck, trial consortia). - Computational resources (GPU clusters, pathology servers).

Impact

This project addresses the urgent need for robust PD-L1 biomarkers in TNBC. By integrating CNN image classification, protocol imaging, and AI Agent orchestration, CPS++ will provide a reproducible, patentable biomarker with strong clinical translation potential for immunotherapy response prediction.